Software Engineering

The Smart Retail Demand Forecasting System

RetailIQ

Submitted by:

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Submitted to:

Ms. Ansa Liaqut

Date of submission: 20/2/26

Smart Retail Demand Forecasting System (RetailIQ)

1. Project Overview

RetailIQ is a web-based retail analytics platform designed to help businesses forecast product demand and understand the factors influencing sales performance. The system uses historical retail sales data to generate demand forecasts and provide meaningful insights through interactive dashboards and analytical tools.

Traditional retail forecasting tools only predict future sales numbers but fail to explain the reasons behind demand fluctuations. RetailIQ improves this by providing interpretable analysis of sales patterns, promotion impacts, and product relationships. The platform allows users to upload sales datasets, visualize sales trends, analyze promotional effects, and receive business recommendations based on data-driven insights.

The system integrates machine learning models with a user-friendly web interface to help retail managers make informed decisions regarding inventory management, promotions, and product stocking strategies. Through interactive visualizations and analytical dashboards, RetailIQ transforms complex retail data into actionable insights.

2 User Stories

Sprint 1 focuses on data preparation and exploratory analysis. Advanced forecasting and prediction features will be implemented in future sprints

<table>
<tr><td>Story ID: US-01Story Title: Data Loading</td><td>Priority: High
Estimated Hours: 6</td></tr>
<tr><td>User Story:
As a retail analyst, I want to upload and access retail sales data in the system so that I can analyze sales performance easily.
Description:
This story involves enabling users to upload various retail datasets (such as sales, stores, and oil data), validate them, and provide a summary view for quick understanding.
Sub User Stories:
• As a user, I want to upload datasets (sales, stores, oil).
• As a user, I want the system to validate uploaded data.
• As a user, I want to view dataset summaries.
Acceptance Criteria:
• • Users should be able to upload datasets successfully.
• The system should validate the data and show errors if any.
• A summary view of uploaded data should be displayed.</td><td></td></tr>
</table>

Story ID: US-02Story Title: Data Cleaning	**Priority:** High **Estimated Hours:** 7
User Story: As a retail analyst, I want the system to automatically clean and preprocess data so that I can work with accurate and reliable information. **Description:** This story focuses on automating data preprocessing tasks such as handling missing values, detecting outliers, and identifying holidays that impact sales. **Sub User Stories:** • As a user, I want missing values to be handled automatically. • As a user, I want unusual sales values (outliers) to be detected. • As a user, I want holidays affecting sales to be identified. **Acceptance Criteria:** • • • • Missing values should be handled appropriately. • Outliers should be detected and flagged. • Holidays affecting sales should be identified and marked	

<table>
<tr>
<td>Story ID: US-03Story Title: Feature Insights</td>
<td>Priority: Medium

Estimated Hours: 6</td>
</tr>
<tr>
<td>
User Story:

As a business manager, I want the system to generate meaningful features from sales data so that I can understand patterns and trends in business performance.

Description:

This story involves generating analytical insights such as time-based patterns, trends, and rolling averages to help users understand business performance.

Sub User Stories:
• As a user, I want to view monthly and weekly sales patterns.
• As a user, I want to analyze past sales trends.
• As a user, I want to see rolling averages of sales.

Acceptance Criteria:
• The system should display monthly and weekly patterns.
• Historical sales trends should be visualized.
• Rolling averages should be calculated and shown
</td>
<td></td>
</tr>
</table>

Story ID: US-04

Story Title: Data Visualization (EDA)

Priority: HIgh

Estimated Hours: 5

User Story:

As a store owner, I want to see visual reports of sales trends and seasonality so that I can make informed business decisions.

Description:

This story focuses on providing visual exploratory data analysis (EDA) tools, including trend graphs, seasonal patterns, and downloadable reports.

Sub User Stories:

- As a user, I want to view trend graphs.
- As a user, I want to view seasonal patterns.
- As a user, I want to download visual reports.

Acceptance Criteria:

- Trend graphs should be displayed clearly.
- Seasonal patterns should be visualized.
- Users should be able to download visual reports.

3 Structured Specifications

Specification for US-01: Data Loading

Scenario 1: Upload dataset successfully

Given the user is on the data upload page
When the user uploads a valid dataset (sales, stores, or oil)
Then the system should accept the file
And display a success message

Scenario 2: Invalid dataset upload

Given the user is on the data upload page
When the user uploads an invalid or corrupted dataset
Then the system should reject the file
And display an error message indicating the issue

Scenario 3: View dataset summary

Given the user has successfully uploaded a dataset
When the user navigates to the dataset overview section
Then the system should display a summary (rows, columns, basic stats)

Specification for US-02: Data Cleaning

Scenario 1: Handle missing values

Given the dataset contains missing values
When the system performs preprocessing
Then missing values should be handled automatically
And the cleaned dataset should be updated

Scenario 2: Detect outliers

Given the dataset contains unusual sales values
When the system analyzes the data
Then outliers should be detected
And flagged for the user

Scenario 3: Identify holidays

Given the dataset includes date-related sales data
When the system processes the dataset

Then holidays affecting sales should be identified
And marked in the dataset

Specification for US-03: Feature Insights

Scenario 1: View weekly and monthly patterns

Given the dataset has been processed
When the user selects the patterns view
Then the system should display weekly and monthly sales patterns

Scenario 2: Analyze past trends

Given the user is viewing feature insights
When the user selects trend analysis
Then the system should display historical sales trends

Scenario 3: View rolling averages

Given the dataset is available
When the user selects rolling average analysis
Then the system should calculate rolling averages
And display them visually

Specification for US-04: Data Visualization (EDA)

Scenario 1: View trend graphs

Given the user is on the visualization dashboard
When the user selects trend graphs
Then the system should display sales trend graphs

Scenario 2: View seasonal patterns

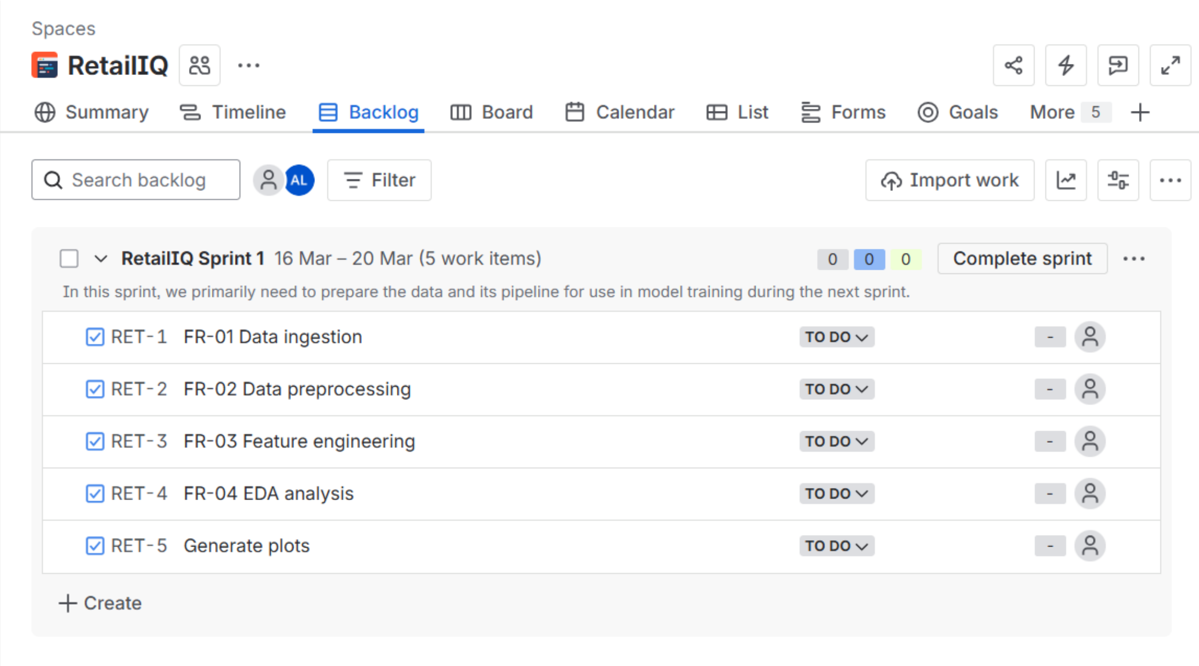
Given the user is on the visualization dashboard
When the user selects seasonal analysis
Then the system should display seasonal patterns in sales

Scenario 3: Download visual reports

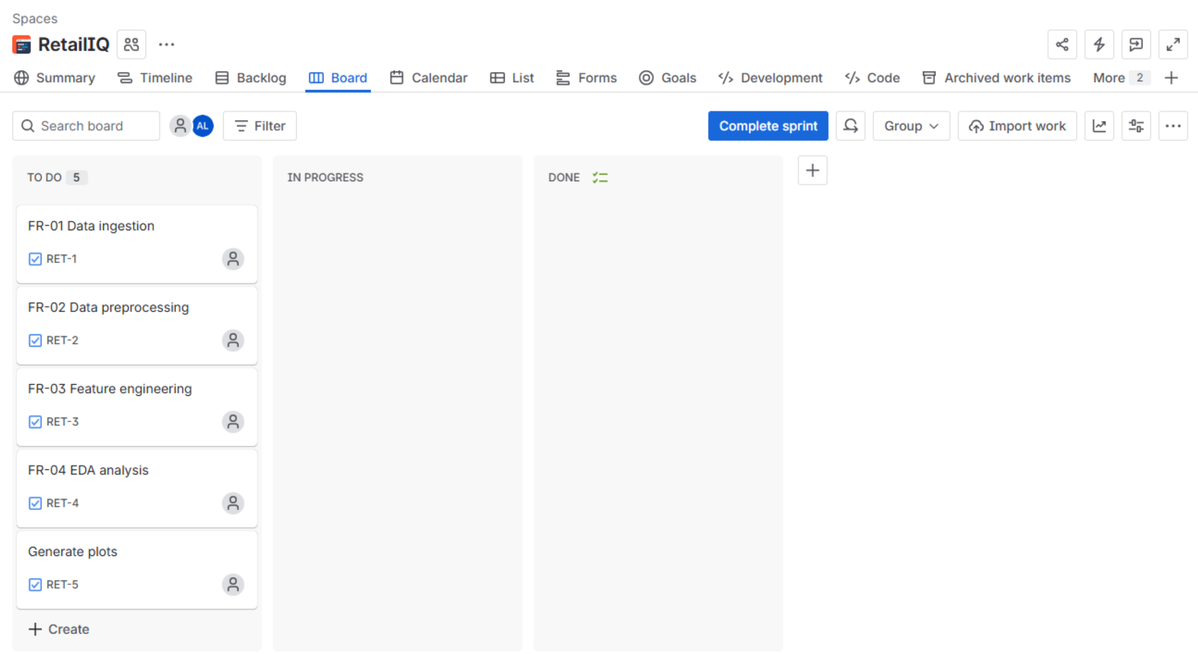
Given the user has generated a visualization
When the user clicks the download option
Then the system should export the report (e.g., PDF/PNG)
And download it successfully

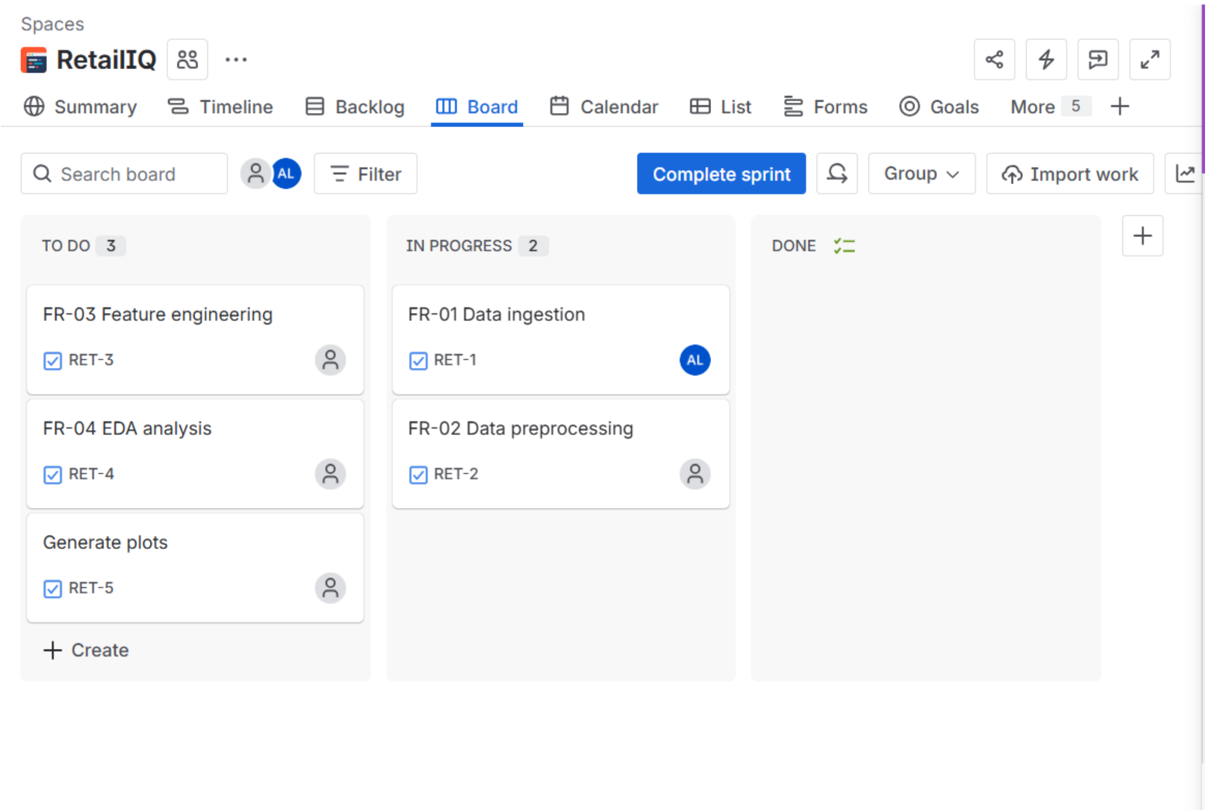
4 Scrum Board

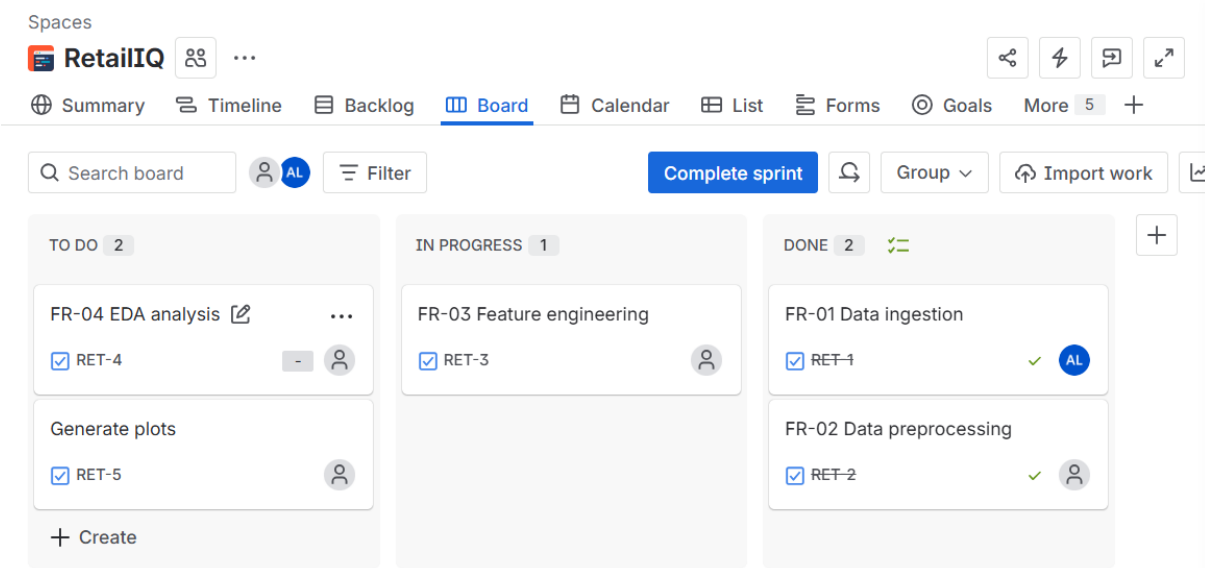
Backlog for Sprint 1:

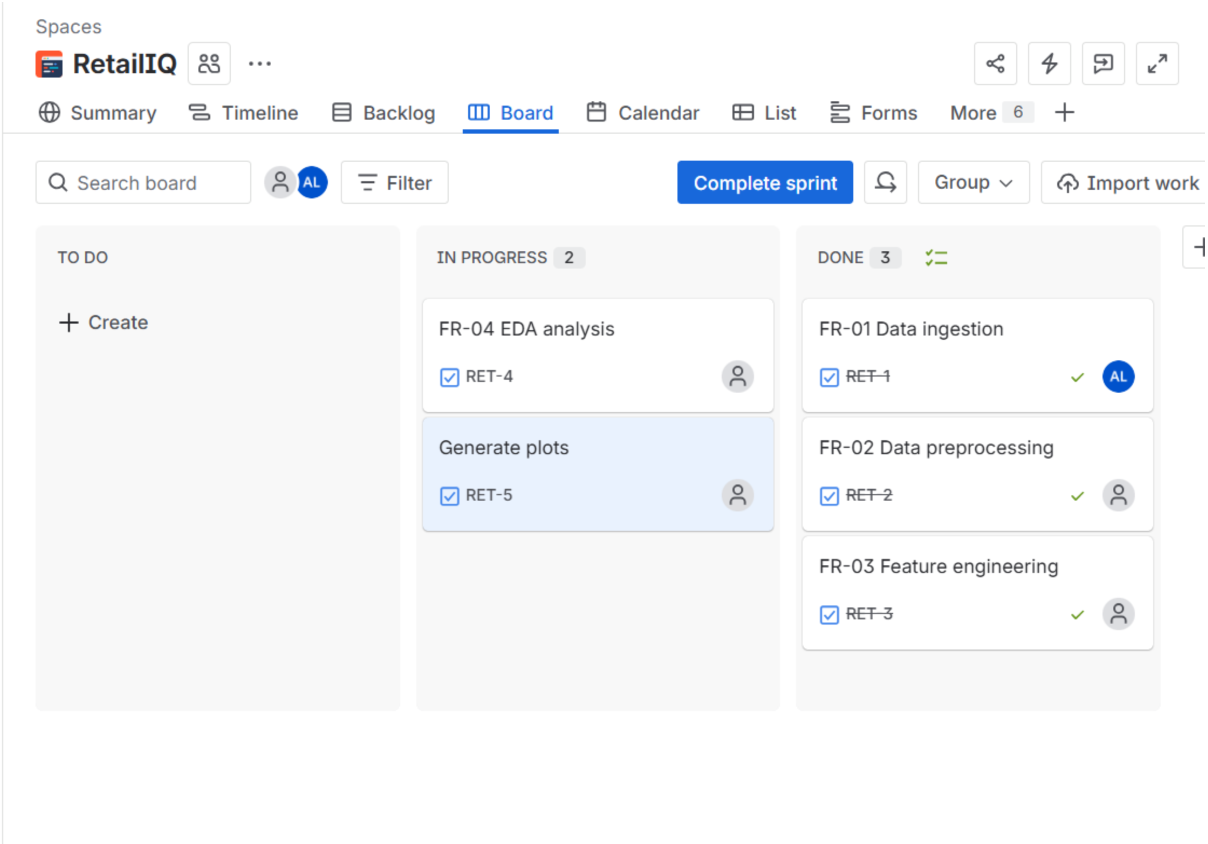


Sprint Board:

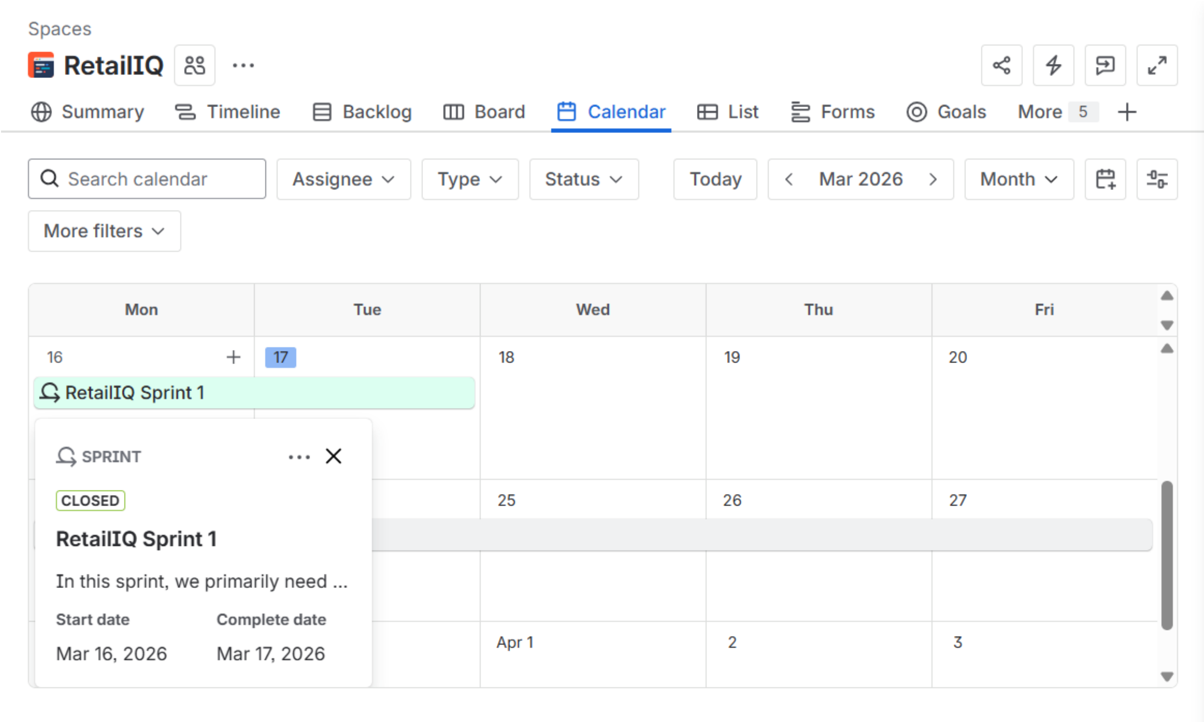



Spaces
RetailIQ
Summary
Timeline
Backlog
Board
Calendar
List
Forms
Goals
More 5
Search board
AL
Filter
Complete sprint
Group
Import work
TO DO 3
IN PROGRESS 2
DONE
FR-03 Feature engineering
RET-3
FR-04 EDA analysis
RET-4
Generate plots
RET-5
Create
FR-01 Data ingestion
RET-1
AL
FR-02 Data preprocessing
RET-2

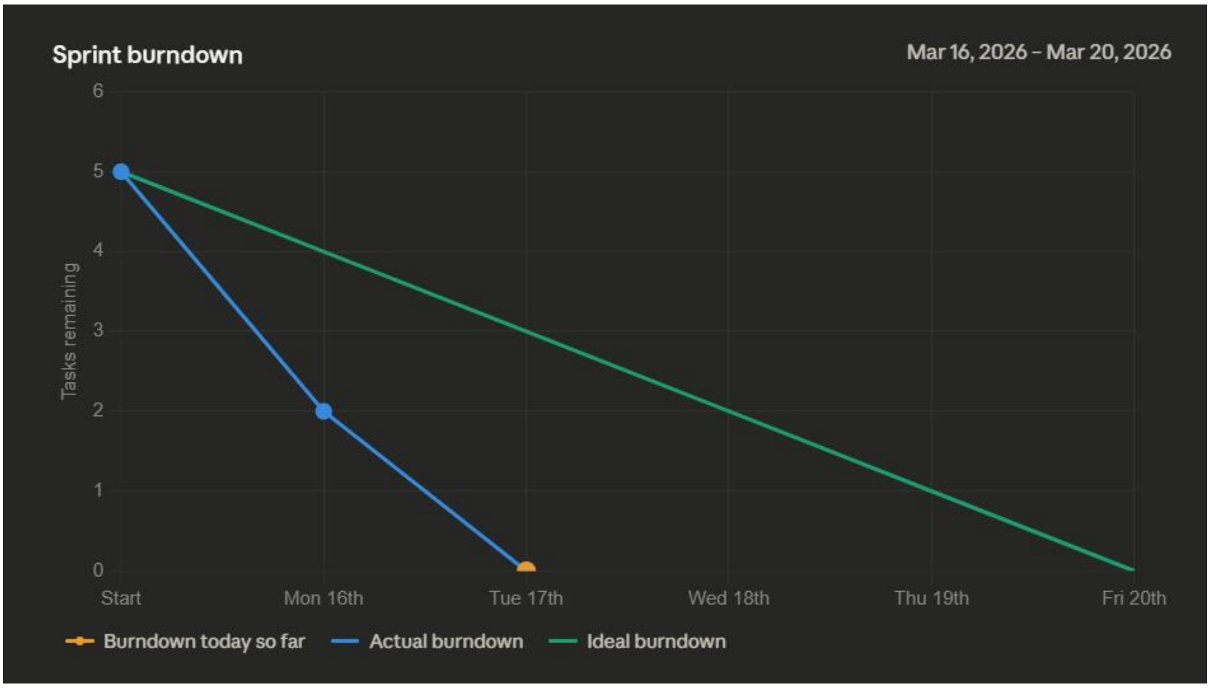

Spaces
RetailIQ
Summary
Timeline
Backlog
Board
Calendar
List
Forms
Goals
More 5
Search board
AL
Filter
Complete sprint
Group
Import work
TO DO 2
IN PROGRESS 1
DONE 2
FR-04 EDA analysis
RET-4
Generate plots
RET-5
Create
FR-03 Feature engineering
RET-3
FR-01 Data ingestion
~~RET-1~~
AL
FR-02 Data preprocessing
~~RET-2~~



Completed Sprint:



Burn Down Chart



5 NFRs for Sprint 1 – Retail Data Module

Your **Sprint 1** only covers **data ingestion, preprocessing, feature engineering, and EDA**, so NFRs should focus on **data handling and usability**.

NFR-01: Performance

The system should load datasets containing millions of sales records within a reasonable time (e.g., <10 seconds) to ensure smooth user experience.

NFR-02: Reliability

The system should handle missing, inconsistent, or corrupted data gracefully without crashing, providing clear error messages to the user.

NFR-03: Usability

The system should allow the user to upload datasets and view summary statistics and visualizations in an intuitive and user-friendly manner.

NFR-04: Scalability

The system should support multiple stores and product families and allow for larger datasets in future sprints without performance degradation.

NFR-05: Maintainability

The data processing and visualization modules should be designed in a modular way so that updates and new features can be added easily in future iterations.

6 Iteration 1 implementation screen shots

Fr-01 Loading Data

```
[3]
train = pd.read_csv("data/train.csv")
test = pd.read_csv("data/test.csv")
stores = pd.read_csv("data/stores.csv")
oil = pd.read_csv("data/oil.csv")
holidays = pd.read_csv("data/holidays_events.csv")
transactions = pd.read_csv("data/transactions.csv")
```
Python

```
[4]
print("train:", train.shape)
print("test:", test.shape)
print("stores:", stores.shape)
print("oil:", oil.shape)
print("holidays:", holidays.shape)
print("transactions:", transactions.shape)
```
Python

```
train: (3000888, 6)
test: (28512, 5)
stores: (54, 5)
oil: (1218, 2)
holidays: (350, 6)
transactions: (83488, 3)
```

```
[30]
null_report = df.isnull().sum().sort_values(ascending=False)

print("Null Count Report:")
print(null_report)
```
Pyth

```
Null Count Report:
locale          2551824
holiday_type    2551824
locale_name     2551824
description     2551824
transferred     2551824
dcoilwtico       928422
transactions     245784
id                    0
date                  0
store_type            0
state                 0
city                  0
onpromotion           0
sales                 0
family                0
store_nbr             0
cluster               0
dtype: int64
```

```
[31]
print("Merged Dataset Shape:", df.shape)
```
Python

```
Merged Dataset Shape: (3000888, 17)
```

```
[10]
stores_count = df['store_nbr'].nunique()
family_count = df['family'].nunique()
days_count = df['date'].nunique()

expected_rows = stores_count * family_count * days_count

print("Expected Rows:", expected_rows)
print("Actual Rows:", len(df))
```
Python

```
Expected Rows: 3000888
Actual Rows: 3000888
```

FR-02 → Data Preprocessing

Missing oil price values were imputed using a forward-fill strategy to propagate the last known trading value across non-trading days. A backward-fill step was additionally applied to handle missing values at the beginning of the dataset, resulting in 0% null values in the dcoilwtico feature.

```python
df.head()
```

	id	date	sales	onpromotion	city	state	store_type	cluster	dcoilwtico	transactions	holiday_type	locale	locale_name	description	tra
0	0	2013-01-01	0.0	0	Quito	Pichincha	D	13	93.14	NaN	Holiday	National	Ecuador	Primer dia del ano	
1782	1782	2013-01-02	2.0	0	Quito	Pichincha	D	13	93.14	2111.0	NaN	NaN	NaN	NaN	
3564	3564	2013-01-03	3.0	0	Quito	Pichincha	D	13	92.97	1833.0	NaN	NaN	NaN	NaN	
5346	5346	2013-01-04	3.0	0	Quito	Pichincha	D	13	93.12	1863.0	NaN	NaN	NaN	NaN	
7128	7128	2013-01-05	5.0	0	Quito	Pichincha	D	13	93.12	1509.0	Work Day	National	Ecuador	Recupero puente Navidad	

```python
df.head()
```

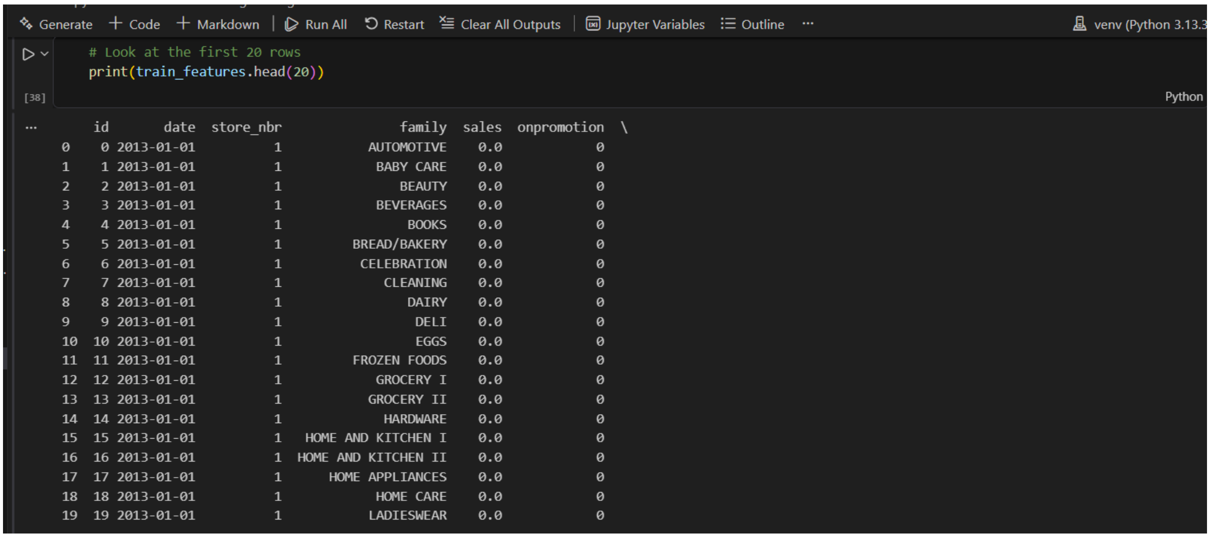
tate	store_type	cluster	dcoilwtico	transactions	holiday_type	locale	locale_name	description	transferred	is_holiday	is_outlier	store_nbr	family
ıcha	D	13	93.14	NaN	Holiday	National	Ecuador	Primer dia del ano	False	True	False	1	AUTOMOTIVE
ıcha	D	13	93.14	2111.0	NaN	NaN	NaN	NaN	NaN	False	False	1	AUTOMOTIVE
ıcha	D	13	92.97	1833.0	NaN	NaN	NaN	NaN	NaN	False	False	1	AUTOMOTIVE
ıcha	D	13	93.12	1863.0	NaN	NaN	NaN	NaN	NaN	False	False	1	AUTOMOTIVE
ıcha	D	13	93.12	1509.0	Work Day	National	Ecuador	Recupero puente Navidad	False	True	False	1	AUTOMOTIVE

```python
print("Oil price nulls:", df['dcoilwtico'].isnull().sum())
print("Holiday flag counts:")
print(df['is_holiday'].value_counts())

print("Outliers detected:", df['is_outlier'].sum())

print("Dataset shape:", df.shape)
```

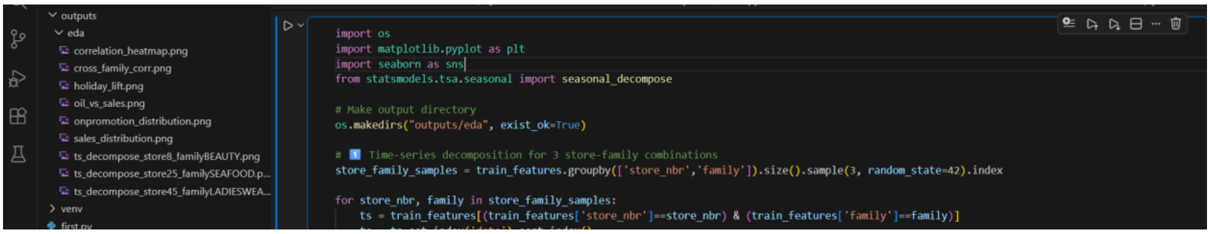
```
Oil price nulls: 0
Holiday flag counts:
is_holiday
False    2567862
True      433026
Name: count, dtype: int64
Outliers detected: 113422
Dataset shape: (3000888, 19)
```

Conclusion: The dataset is now fully cleaned, flagged, and temporally ordered, ready for feature engineering (FR-03).

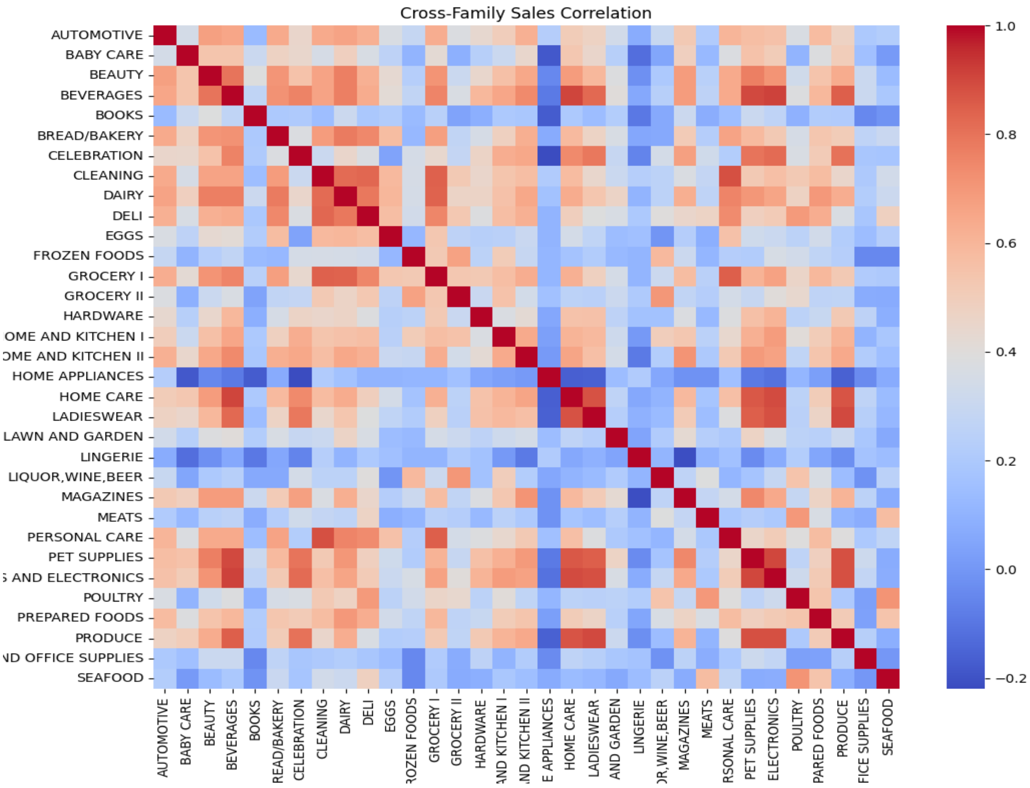
FR-03 Feature Engineering Module



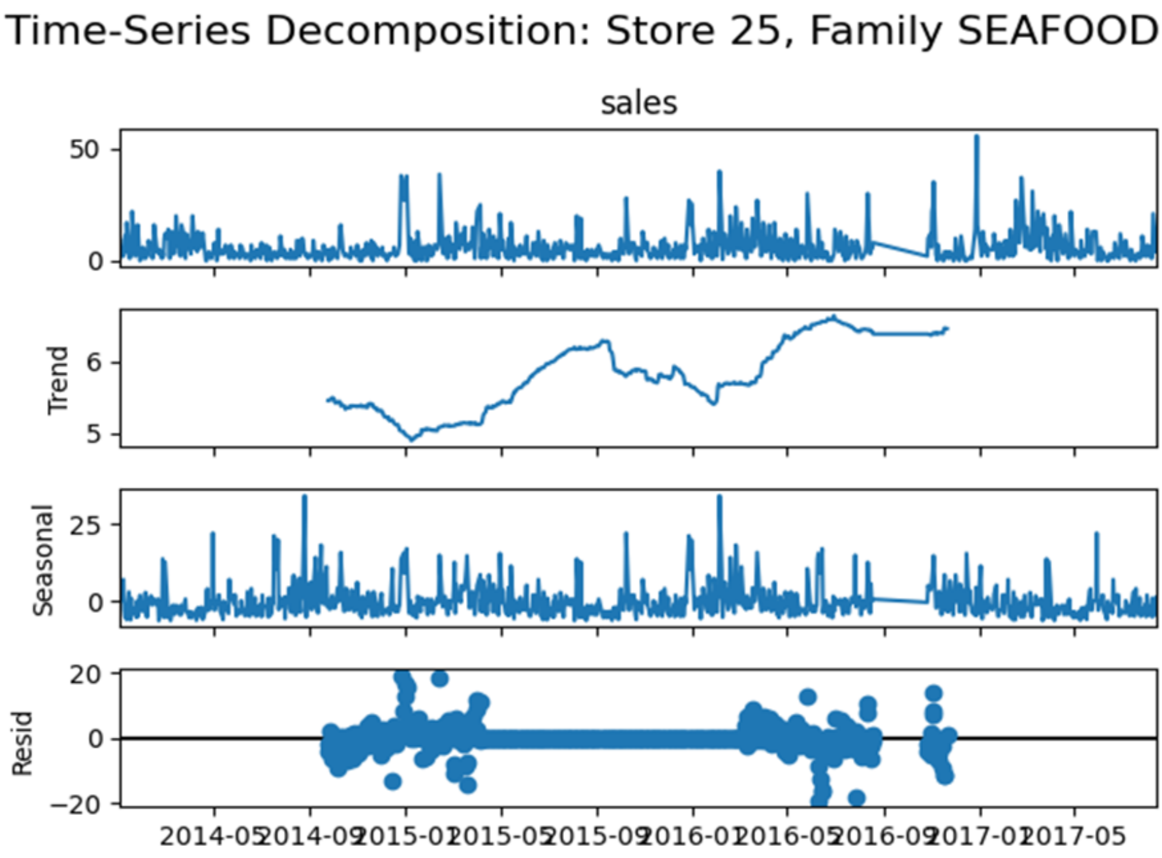
FR-04 EDA



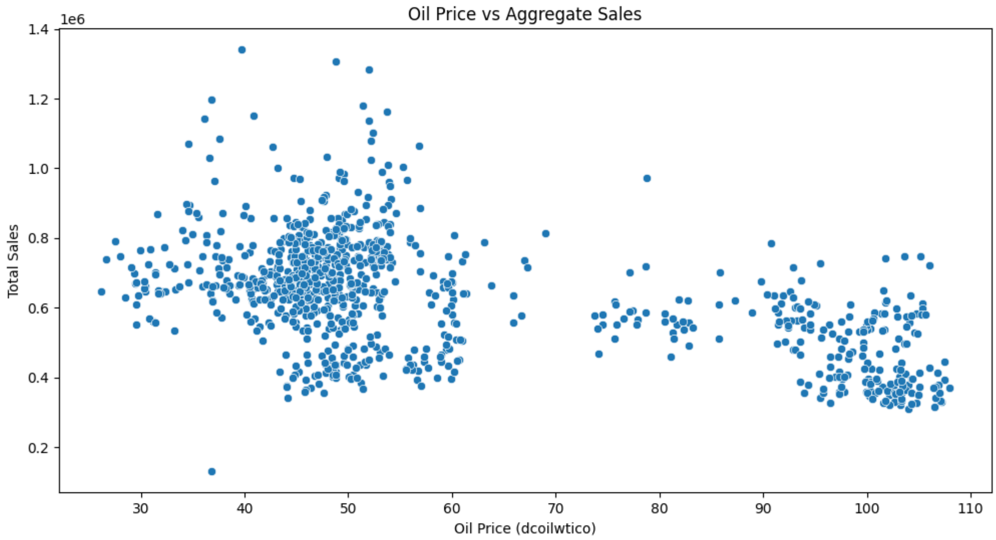
Outputs:



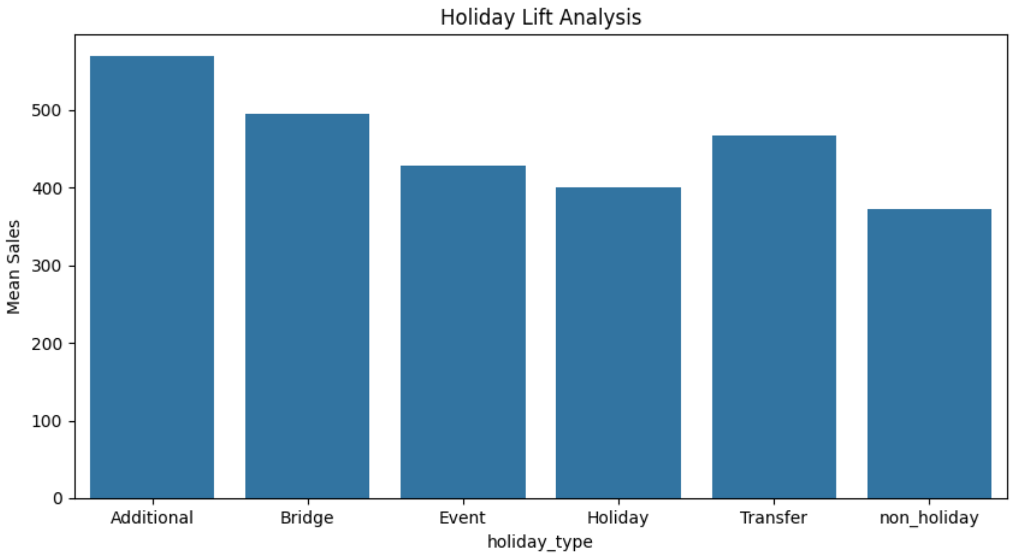
Cross-Family Sales Correlation
AUTOMOTIVE
BABY CARE
BEAUTY
BEVERAGES
BOOKS
BREAD/BAKERY
CELEBRATION
CLEANING
DAIRY
DELI
EGGS
FROZEN FOODS
GROCERY I
GROCERY II
HARDWARE
OME AND KITCHEN I
OME AND KITCHEN II
HOME APPLIANCES
HOME CARE
LADIESWEAR
LAWN AND GARDEN
LINGERIE
LIQUOR,WINE,BEER
MAGAZINES
MEATS
PERSONAL CARE
PET SUPPLIES
; AND ELECTRONICS
POULTRY
PREPARED FOODS
PRODUCE
ND OFFICE SUPPLIES
SEAFOOD
1.0
0.8
0.6
0.4
0.2
0.0
−0.2



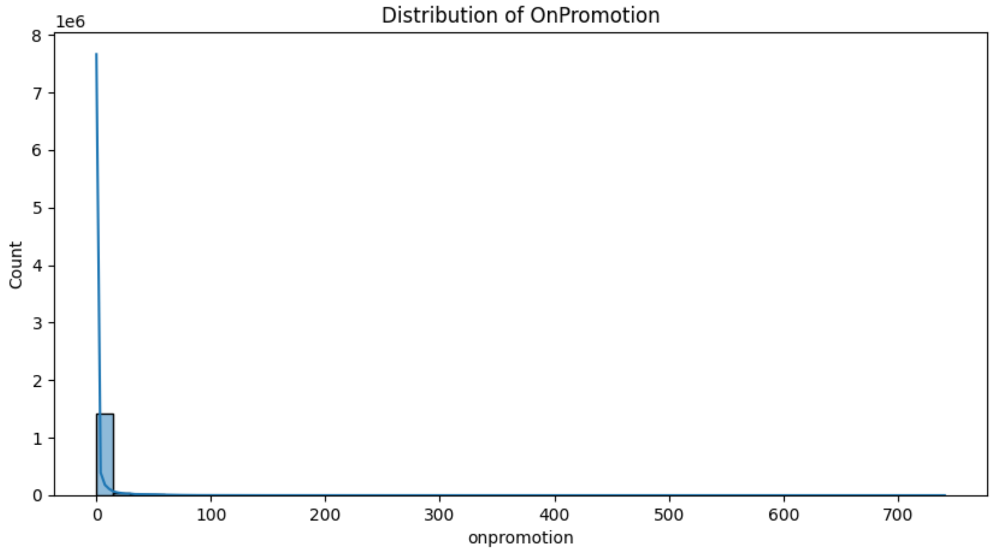
Time-Series Decomposition: Store 25, Family SEAFOOD
sales
Trend
Seasonal
Resid
2014-05
2014-09
2015-01
2015-05
2015-09
2016-01
2016-05
2016-09
2017-01
2017-05

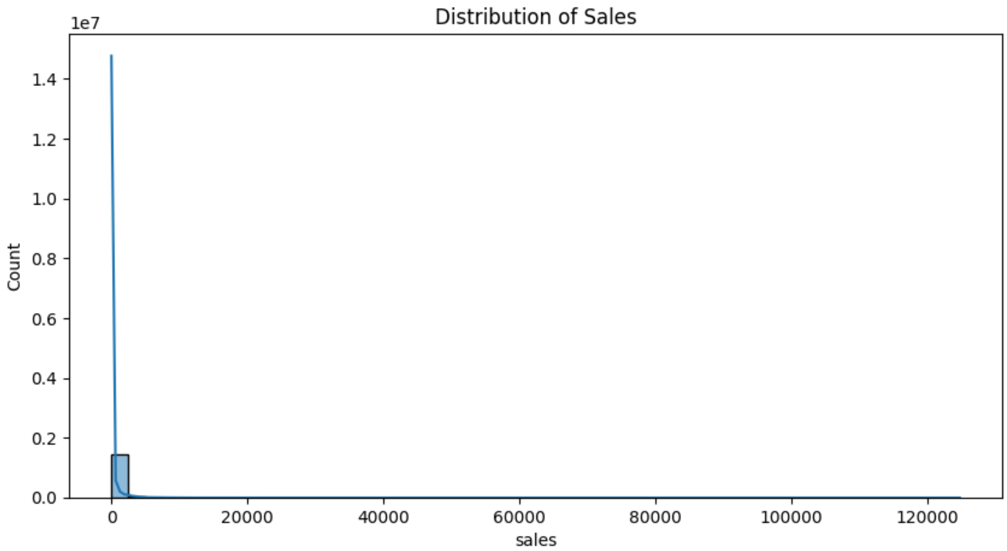


Oil Price vs Aggregate Sales
1e6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
Total Sales
30
40
50
60
70
80
90
100
110
Oil Price (dcoilwtico)



Holiday Lift Analysis
500
400
300
200
100
0
Mean Sales
Additional
Bridge
Event
Holiday
Transfer
non_holiday
holiday_type


Distribution of OnPromotion
1e6
Count
onpromotion
0
100
200
300
400
500
600
700
8
7
6
5
4
3
2
1
0


Distribution of Sales
1e7
Count
sales
0
20000
40000
60000
80000
100000
120000
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0

7:Work Division

Task ID	Task Description	Primary Responsibility
FR-01	**Data Ingestion &Loading**: Developing the module to upload multiple datasets (sales, stores, oil, etc.) and generate summary statistics.	**Ayesha Khalid**
FR-02	**Data Preprocessing &Cleaning**: Implementing logic for handling missing values (forward/backward fill), detecting outliers, and identifying holidays.	**Uma E Rubab**
FR-03	**Feature Engineering**: Creating time-based features, rolling averages, and historical sales patterns to generate business insights.	**Ayesha Khalid**
FR-04	**Exploratory Data Analysis (EDA)**: Developing visual dashboards for trend analysis, seasonal patterns, and interactive visualizations.	**Uma E Rubab**
NFR	**Non-Functional Requirements**: Defining and ensuring system performance, reliability, and usability standards.	**Joint Effort**
DOC	**Documentation & Scrum Management**: Drafting user stories, structured specifications, and maintaining the sprint backlog/board.	**Joint Effort**